

Understanding IPv4 and Related Concepts

IPv4 Address

an **IPv4 address** is a **32-bit logical address** that uniquely identifies a device on a network.

- It's divided into **four sections** (called **octets**), separated by dots (e.g., 192.168.1.1).
- Each octet is 8 bits, which is why they are **32 bits total** ($4 \times 8 = 32$).
- The address is divided into the **Network part** (which identifies the specific network) and the **Host part** (which identifies the specific device on that network).
- The original system classified these addresses into **Class A, B, and C** based on their ranges, determining which bits were for the Network and which were for the Host.

Subnet

A **subnet** (short for **sub-network**) is a smaller, logical division of a larger network.

- **Imagine a large office building** 🏢 (your main network). Instead of having everyone in one giant, chaotic open space, you divide it into different departments (Accounting, HR, Marketing). Each department is a **subnet**.

Use of Subnetting

The main reasons for creating subnets are:

1. **Efficiency:** It reduces the amount of network traffic that every device has to see. Traffic for the Accounting department stays mostly within the Accounting subnet.
2. **Security:** You can easily apply different security policies to different subnets (e.g., restrict Marketing from accessing Accounting's sensitive servers).
3. **Organization:** It helps manage a large number of devices by grouping them logically and physically.

Subnet Mask

The **subnet mask** is a 32-bit number used to tell the network device which part of an IP address is the **Network ID** and which part is the **Host ID**.

- **Network bits** are represented by $\mathbf{1}$ s in the mask.
- **Host bits** are represented by $\mathbf{0}$ s in the mask.

IP Address	192.	168.	1.	10	(Example IP)
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Subnet Mask	255.	255.	255.	0	(Example Mask)
Meaning	Network	Network	Network	Host	

In the example above, the first three octets (192.168.1) are the Network ID, and the last octet (10) is the Host ID.

CIDR Range (Classless Inter-Domain Routing)

CIDR is the modern, more flexible way to define the Network part of an IP address, replacing the old Class A/B/C system.

- It's written as an IP address followed by a **forward slash** and a number, like **192.168.1.0/24**.
- The number (e.g., **24**) is the **CIDR value**, which simply tells you **exactly how many bits** (starting from the left) are being used for the **Network ID** (the $\mathbf{1}$ s in the subnet mask).
- **Example:** A **/24** means the first 24 bits are the Network ID, which is the same as a mask of **255.255.255.0**.

Network ID (N/W ID)

The **Network ID** (or **Network Address**) is the **first address** in a subnet range.

- It is the address where **all the Host bits are $\mathbf{0}$** .
- This address is used to **identify the entire network** and **cannot be assigned to a device**.

Broadcast ID

The **Broadcast ID** is the **last address** in a subnet range.

- It is the address where **all the Host bits are $\mathbf{1}$** .
- A message sent to this address reaches **every device** on that specific subnet.
- Like the Network ID, it **cannot be assigned to a device**.

Subnetting Example Calculation

Let's use an example to see how it all works!

Scenario

You are given the **main network 192.168.1.0/24** and need to create **8 subnets** for 8 different office departments.

Step 1: Determine the Number of Borrowed Bits (\$n\$)

You need 8 subnets. The formula for the number of possible subnets is 2^n , where n is the number of **bits you borrow from the Host part**.

$$2^n$$

Since $2^3 = 8$, you need to **borrow $n=3$ bits**.

Step 2: Calculate the New Subnet Mask (CIDR)

- The original network was **/24** (24 Network bits).
- You borrowed **3** bits for the network.
- **New CIDR:** $24 + 3 = \mathbf{27}$
- The new subnet mask is **/27**.

Binary Breakdown:

- A **/24** mask in binary is: $\underbrace{11111111.11111111.11111111}_{\text{24 Network Bits}}.\underbrace{00000000}_{\text{8 Host Bits}}$
- A **/27** mask is: $\underbrace{11111111.11111111.11111111}_{\text{27 Network Bits}}.\underbrace{00000}_{\text{5 New Host Bits}}$

The new subnet mask in dotted-decimal form is **255.255.255.224** (since 11100000 in the last octet is $128+64+32 = 224$).

Step 3: Calculate the Number of Hosts per Subnet

The number of remaining **Host bits** is the total number of bits (32) minus the new Network bits (27): $32 - 27 = \mathbf{5}$ Host bits.

- The formula for the number of available hosts is $2^h - 2$, where h is the number of Host bits.
- $2^5 - 2 = 32 - 2 = \mathbf{30}$ **usable hosts** per subnet.

Step 4: Calculate the Subnets (Network IDs and Broadcast IDs)

The difference between one Network ID and the next is called the **block size** or **increment**. This is calculated using the last octet of the new mask (224) by subtracting it from 256:

- **Block Size:** $256 - 224 = \mathbf{32}$.

This block size tells you how the subnets increment.

Subnet Number	Network ID (First Address, Host bits are 0)	Usable Range (First Host - Last Host)	Broadcast ID (Last Address, Host bits are 1)
1	192.168.1.0/27	192.168.1.1 - 192.168.1.30	192.168.1.31
2	192.168.1.32/27	192.168.1.33 - 192.168.1.62	192.168.1.63
3	192.168.1.64/27	192.168.1.65 - 192.168.1.94	192.168.1.95
...	<i>Increments by 32</i>		<i>Next Network ID minus 1</i>

While modern networking primarily uses **CIDR (Classless Inter-Domain Routing)** for efficiency, the classes provide a foundational understanding of how network and host bits are structured.

IPv4 Address Classes and Ranges

The class of an IPv4 address is determined by the value of the **first octet** (the first number) of the address. The original system defined five classes: A, B, C, D, and E.

Key Points for A, B, and C (Unicast Classes)

The first three classes (A, B, and C) are known as **Unicast** classes, meaning they are used for communication between a single source and a single destination.

- **Class A (/8):** Uses the **first octet** for the Network ID and the remaining **three octets** (24 bits) for Host IDs. This results in **few networks** but an extremely **large number of hosts** per network ($2^{24} - 2$ approx 16.7\$ million hosts).

Class	First Octet Range	Total Range (First Octet)	Default Subnet Mask	CID R	Network/Host Split	Purpose
A	1 to 126	1.0.0.0 to 126.255.255.255	255.0.0.0	8	Network .Host.Host.Host	Very large networks (e.g., massive corporations, ISPs).
B	128 to 191	128.0.0.0 to 191.255.255.255	255.255.0.0	16	Network.Network .Host.Host	Medium to large networks (e.g., major universities, large companies).
C	192 to 223	192.0.0.0 to 223.255.255.255	255.255.255.0	24	Network.Network.Network .Host	Small networks (e.g., small businesses, home networks).
D	224 to 239	224.0.0.0 to 239.255.255.255	N/A	N/A	N/A	Multicast (used for sending data to a group of devices)

						simultaneously).
E	240 to 255	240.0.0.0 to 255.255.255.255	N/A	N/A	N/A	Experimental (Reserved for future or research purposes).

- *Reserved Example:* The range \$127.0.0.0\$ to \$127.255.255.255\$ is reserved for the **loopback address** (like \$127.0.0.1\$) used for local host testing.
- **Class B (/16):** Uses the **first two octets** for the Network ID and the remaining **two octets** (16 bits) for Host IDs. This offers a **moderate number of networks** and a **moderate number of hosts** per network ($2^{16} - 2 \approx 65,000$ hosts).
- **Class C (/24):** Uses the **first three octets** for the Network ID and only the **last octet** (8 bits) for Host IDs. This results in a **very large number of networks** but a **small number of hosts** per network ($2^8 - 2 = 254$ hosts).

Special Reserved Private Ranges

Within the Class A, B, and C ranges, specific address blocks are reserved for use in **private networks** (like your home or office). These are non-routable on the public internet, which helps conserve the public IPv4 address space.

- **Class A Private:** 10.0.0.0 to 10.255.255.255
- **Class B Private:** 172.16.0.0 to 172.31.255.255
- **Class C Private:** 192.168.0.0 to 192.168.255.255 (The most common range you see on home routers).

Why Classes Are Mostly Obsolete

The fixed nature of classful addressing led to massive **wasted IP addresses**. For example, a medium-sized company needing 1,000 hosts would have to take a whole Class B network (getting over 65,000 host addresses), leaving over 64,000 addresses unused.

This inefficiency led to the development of **CIDR (Classless Inter-Domain Routing)**, which you mentioned, where the network boundary is flexible (e.g., /27, /22, /19) and determined only by the subnet mask, not the first octet. This allows for much more efficient allocation of IP addresses.

